

TuniTech Abroad

Product Requirements Document (PRD)

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1. Product Summary

TuniTech Abroad is a France-first job intelligence platform helping Tunisian IT profiles find realistic France IT jobs or internships, compare their CV/profile against job requirements, detect missing skills, and receive practical guidance before applying.

The product is not a generic job board. It combines job aggregation, CV/profile parsing, canonical skill matching, recommendations, and controlled AI explanations.

2. MVP Goal

Validate whether Tunisian IT candidates, students, bootcamp graduates, and internship seekers will use a platform that gives practical job-fit intelligence for France opportunities.

1. Find relevant France IT jobs and internships.
2. Upload or complete a CV/profile.
3. Receive ranked recommendations.
4. Understand match score and missing skills.
5. Save or apply to better-targeted opportunities.

3. Product Scope

3.1 Included in MVP

- French UI and French email templates.
- Landing page, public job search, job detail pages.
- France Travail API ingestion as primary job source.
- Raw job records and normalized jobs.
- Google, GitHub, and email/password authentication.
- Email verification and password reset.
- Anonymous quick-match teaser.
- PDF CV upload and parsing.
- Manual profile completion.
- Flat canonical skill taxonomy with aliases.
- Rule-based candidate-job fit scoring.
- Recommendation ranking and stored recommendations.
- Controlled OpenRouter LLM explanations and extraction support.

- Saved jobs, match history, email preferences, weekly digest.
- Django Admin dashboard and operational monitoring.
- Privacy basics: consent, policies, delete CV, delete account.

3.2 Excluded from MVP

- Employer/recruiter/training-center dashboards.
- Payments/subscriptions.
- Belgium/Luxembourg/Canada/Switzerland jobs.
- LinkedIn profile import, LinkedIn login, Facebook login.
- Auto-apply, chatbot, mobile app, salary prediction, company reviews, messaging.
- AI-only scoring or bulk LLM ranking.
- Legal immigration advice or guaranteed employment/relocation claims.

4. Personas

Persona	Description	Needs
Junior IT candidate	0-2 years experience; may target first job, junior role, internship, or PFE.	Realistic recommendations, missing skills, CV guidance.
Final-year student	Preparing for PFE, internship, first job, or future France opportunities.	Market visibility, roadmap, internship filtering, CV/project positioning.
Bootcamp graduate	Candidate entering IT market through bootcamp path.	Skill gap roadmap, expectations, project-based proof.
Mid-level developer	2-5 years experience targeting France IT roles.	Better targeting, stronger CV alignment, advanced skill gaps.

Personas are planning profiles. They do not require separate site modes in MVP.

5. User Access Rules

User type	Can do	Cannot do
Anonymous visitor	View landing page, search jobs, view job details, run quick-match teaser.	Upload CV, save jobs, receive full recommendations, receive email digest.
Authenticated user	Upload CV, complete profile, generate matches, get recommendations, save jobs, manage emails.	Access other users data or admin features.
Admin user	Monitor users, profiles, CVs, jobs, ingestion, recommendations, LLM usage, emails.	Access production secrets from UI.

6. Core User Journeys

6.1 Job-First Full Match

6. User lands on homepage.
7. User searches France IT jobs or internships.
8. User opens job detail page.

9. User clicks Check my match.
10. User signs up/logs in.
11. User uploads CV.
12. System extracts profile data.
13. User confirms and completes missing fields.
14. System calculates fit score and explanation.
15. User saves job or opens source application link.

6.2 Anonymous Quick-Match Teaser

16. Anonymous user opens a job.
17. User enters 3-5 skills, experience level, and French level.
18. System returns estimated teaser score, matched skills, and likely missing skills.
19. User is prompted to create an account for full CV analysis and recommendations.

6.3 Profile-First Recommendations

20. User signs up/logs in.
21. User uploads CV.
22. System extracts profile data.
23. User completes profile.
24. System computes recommended France jobs/internships.
25. Dashboard shows ranked recommendations and missing skills.

7. Public Pages

Page	Requirements
/	French landing page with product headline, problem, how it works, target users, France-first scope, CTAs.
/jobs	Public job search from local DB with filters and active jobs only by default.
/jobs/[id]	Job detail page with title, company, location, contract, description, skills, freshness, source link, quick-match/full-match CTAs.
/cv-checker	Explains CV upload, profile extraction, recommendations, and missing skill analysis; login needed for full upload.
/login	Google, GitHub, email/password login.
/register	Google, GitHub, email/password registration with verification for email signup.
/privacy	Privacy Policy page.
/terms	Terms page.

8. Private Dashboard Pages

Page	Requirements
/dashboard	Profile completeness, CV status, top recommendations, top missing skills, saved jobs, match history, email digest status.
/dashboard/profile	Editable profile: phone, location, LinkedIn, GitHub,

	portfolio, target roles, target type, languages, preferences.
/dashboard/cv	Upload/replace active PDF CV, parse status, extracted fields, warnings, delete CV.
/dashboard/recommendations	Ranked jobs with fit score, missing skills, risk flags, filters, freshness.
/dashboard/matches	Historical match analyses with score breakdown and details.
/dashboard/saved-jobs	Saved jobs with active/stale/expired labels and remove action.
/dashboard/email-preferences	Weekly digest and product update preferences.
/dashboard/account	Account settings, delete account/data action.

9. Authentication Requirements

- Support Google OAuth, GitHub OAuth, and email/password.
- Email/password users must verify email.
- OAuth users must still complete profile fields.
- If the same verified email appears across providers, the system should avoid duplicate accounts where safely possible.
- LinkedIn and Facebook are excluded from MVP.

10. Profile Requirements

10.1 Profile Fields

- Full name, email, phone, location.
- LinkedIn URL, GitHub URL, portfolio URL, personal website.
- Current level: Student, Internship seeker, Junior, Mid-level, Senior.
- Years of experience.
- Target roles.
- Target country: France for MVP.
- Target type: summer internship, standard internship, PFE/end-of-study internship, first job, junior job, mid-level job.
- French level and English level.
- Main technologies and skills.
- Relocation preference and remote preference.

10.2 Completeness Score

Field group	Weight
Active CV uploaded	20
Main skills	20
Target role/type	15
Experience level/years	15
French/English level	10
Contact/profile links	10
Location/relocation/remote preference	10

Completeness labels: below 50 = incomplete; 50-79 = usable; 80+ = strong.

11. CV Upload and Parsing Requirements

- MVP accepts PDF only.
- User has one active CV and may replace it.
- Replacing CV triggers parsing and recommendation refresh.
- User may delete CV.
- System must require explicit consent before processing CV.

CV parsing should attempt to extract name, email, phone, location, LinkedIn, GitHub, portfolio, website, skills, work experience, projects, education, certifications, languages, and estimated years of experience.

Parse statuses: pending, processing, parsed, parsed_with_warnings, failed.

12. Job Ingestion Requirements

- Primary source: France Travail API Offres d'emploi.
- Future alternance source: La Bonne Alternance API if needed.
- External API is used by scheduled ingestion, not per user search.
- Store raw payloads and normalized jobs.
- Track first_seen_at, last_seen_at, last_fetched_at, source status, and normalized status.
- Ingestion every 3-6 hours; stale/expiry check every 6-12 hours; full cleanup daily.
- High-intent revalidation can refresh a specific old job on detail/apply click.

12.1 Job Statuses

- active
- stale
- expired
- removed
- archived

Public search shows active jobs by default. Saved jobs may show stale/expired labels.

13. Normalized Job Requirements

- Source name and source job ID.
- Title, company, location, country.
- Contract type, remote type, internship/job type.
- Description and source URL.
- Published date, expiry date if available, first seen, last seen, last fetched.
- Required skills, optional skills, experience level, language requirements.
- Status and normalization/skill extraction status.

Normalization must tolerate missing source fields. Raw payload remains the source audit trail.

14. Job Search Requirements

- Search local normalized jobs only.
- Filters: keyword, role, skill, location, contract type, internship/job type, remote type, experience level, date posted.
- Sort options: newest, most relevant, recently seen, best match for logged-in users with profile.
- Job result card shows title, company, location, contract, internship/job label, remote type, freshness, extracted skills, and View job CTA.

15. Skill Taxonomy Requirements

MVP uses a flat canonical skill taxonomy with aliases. Matching must use canonical skills, not raw extracted strings.

Model	Fields / purpose
Skill	canonical_name, category, is_active.
SkillAlias	skill, alias, normalized_alias.
UnknownSkillCandidate	raw skill text, source type, source ID, first seen, review status.

- Start with 200-300 canonical IT skills.
- Use 500-1000 aliases.
- Unknown extracted skills go to admin review.
- LLM can suggest skills but cannot bypass canonical mapping for scoring.

16. Matching Requirements

Final candidate-job fit score must be deterministic and explainable. LLM may support extraction or explanation but is not the scoring authority.

Fit score component	Weight
Technical skills match	45%
Experience level match	20%
Role/title match	15%
Language fit	10%
Location/remote/relocation fit	10%

16.1 Match Output

- Overall fit score and score breakdown.
- Strong matched skills.
- Missing required and optional skills.
- Experience fit, language fit, location/remote fit.
- Risk flags and recommended actions.

16.2 Risk Flags and Profile Signals

- Risk flags: experience_too_low, missing_required_skills, french_level_missing, english_level_missing, profile_incomplete, job_may_be_expired, internship_type_unclear.

- Profile improvement signals: missing_github, missing_linkedin, missing_portfolio. These do not reduce fit score unless the job explicitly requires them.

16.3 Score Labels

Score range	Label
80-100	Strong match
65-79	Good potential
50-64	Partial match
Below 50	Weak match

17. Recommendation Requirements

Recommendations are included in MVP and must be stored instead of recomputed on every page load.

17.1 Active User Definition

Active user = logged in within the last 14 days and has a usable profile with an active CV or enough manual profile data.

17.2 Recommendation Triggers

- CV uploaded or replaced.
- Profile updated.
- New jobs imported.
- Dashboard opened and recommendations are stale.
- Nightly refresh for active users.

17.3 Recommendation Algorithm

26. Get user profile and active CV/profile version.
27. Pre-filter active jobs by target role, target type, skill overlap, location/remote preferences.
28. Compute deterministic fit score for candidate jobs.
29. Apply ranking score using fit score plus freshness/user preference boosts and stale-job penalty.
30. Store top recommendations with score breakdown and reasons.

Recommendation is stale if computed_at is older than 24 hours, the user profile changed, or the user CV changed.

18. AI / LLM Requirements

- Provider: OpenRouter.
- Allowed: CV extraction assistance, job skill extraction assistance, match explanation, CV improvement suggestions, missing skill roadmap.
- Forbidden: final score alone, bulk ranking all jobs, every search result, no-cache calls, employment guarantees, legal immigration advice.
- Cache CV analysis by cv_hash + prompt_version.
- Cache job analysis by source_job_id + job_updated_at + prompt_version.
- Cache match explanation by cv_hash + job_id + prompt_version.

- Track LLM usage logs with prompt type, model/provider, token/cost estimate, cache hit/miss, errors.

19. Email Requirements

- Required transactional emails: email verification, password reset.
- Optional MVP transactional email: CV analysis completed.
- Weekly recommendation digest is opt-in only.
- Instant high-match alerts are later.
- Digest sends only when user email is verified, opted in, active within last 14 days, has usable profile, and has new relevant recommendations.

20. Saved Jobs Requirements

- Authenticated users can save jobs.
- Saved jobs store user, job, saved date, and status.
- Expired/stale jobs remain visible with warning labels.
- User can remove saved jobs.

21. Privacy and Data Rights Requirements

- Privacy Policy page and Terms page are required.
- CV upload requires explicit consent checkbox.
- User can delete CV.
- User can delete account and associated CV/profile/match/recommendation data.
- Recommendation emails require opt-in and unsubscribe/preferences link.
- CV files are never public.
- Admin access to CVs and personal data is restricted to staff.
- Platform must avoid employment, relocation, visa, or legal guarantees.

22. Admin Dashboard Requirements

Django Admin is acceptable for MVP. Admin must support monitoring, not only CRUD.

- Users, social accounts, email verification status, last login.
- Profiles, completeness, target role/type, skills, language levels.
- CV uploads, parse status, errors, extracted data.
- Job sources, raw job records, normalized jobs, freshness/status.
- Skill taxonomy, aliases, unknown skill candidates.
- Ingestion runs: fetched, created, updated, expired/stale, errors.
- Match results and recommendations.
- LLM usage logs and cost/cache indicators.
- Email events and delivery failures.

23. Data Entities

- User

- SocialAccount
- Profile
- CVUpload
- CVParsedData
- JobSource
- RawJobRecord
- NormalizedJob
- Skill
- SkillAlias
- UnknownSkillCandidate
- JobSkill
- MatchResult
- Recommendation
- SavedJob
- EmailPreference
- EmailEvent
- LLMUsageLog
- IngestionRun
- AccountDeletionRequest or deletion audit record if needed.

24. Non-Functional Requirements

Area	Requirement
Performance	Public job search must load quickly; dashboard must not wait on long recommendations or LLM calls.
Reliability	Ingestion, parsing, email, and LLM failures must be logged and visible in admin.
Security	CV files private; admin restricted; OAuth secrets in env; uploaded files validated; user data isolated.
Privacy	Consent, deletion, no public CVs, opt-in emails, no unnecessary data collection.
Scalability	Handle thousands of jobs, hundreds/low-thousands of users, background jobs, cached LLM outputs.
Maintainability	Scoring logic testable, score breakdown stored, prompt versions tracked, normalization source-specific.

25. Error and Empty States

State	Message / behavior
No jobs found	No matching jobs found. Try removing filters or searching another skill.
CV parse failed	We could not extract enough data from your CV. You can still complete your profile manually.
Profile incomplete	Complete your profile to improve recommendations.
Recommendations pending	Your recommendations are being prepared. Existing jobs can still be searched manually.
Job expired	This job may no longer be active. Application availability is not guaranteed.

26. MVP Acceptance Criteria

26.1 Public Website

- Landing page works.
- Public job search works from local database.
- Job detail page works.
- Anonymous quick-match teaser works.

26.2 Authentication

- Email/password signup and verification work.
- Google login works.
- GitHub login works.
- Logout works.
- Duplicate account risk is handled by email matching where safe.

26.3 CV/Profile

- User can upload PDF CV with consent.
- System extracts basic data.
- User can complete/edit profile.
- Completeness score works.
- User can replace and delete CV.

26.4 Jobs

- France Travail ingestion imports jobs.
- Raw records stored.
- Normalized jobs stored.
- Job freshness statuses tracked.
- Expired/stale handling works.

26.5 Matching and Recommendations

- User can generate match score.
- Score breakdown shown.
- Strong/missing skills shown.
- Recommendations generated and stored.
- Recommendations refresh after CV/profile change.

26.6 AI and Email

- LLM explanations are optional and cached.
- LLM usage is logged.
- Verification/password reset emails work.
- Weekly digest can be enabled/disabled and is not sent without opt-in.

26.7 Admin and Privacy

- Admin can monitor users, CVs, jobs, ingestion, skills, matches, recommendations, LLM, email events.
- Privacy Policy and Terms exist.
- Delete CV and delete account/data actions work.

27. Implementation Priority

Phase	Build focus
1. Foundation	Django project, PostgreSQL, auth, layout, landing page, profile, admin.
2. Job Data	France Travail integration, raw/normalized jobs, ingestion worker, search, job detail.
3. CV/Profile	CV upload, PDF parsing, extracted preview, manual profile completion, completeness score.
4. Skill Taxonomy	Canonical skills, aliases, unknown skill review, CV/job skill normalization.
5. Matching	Rule-based fit score, match result page, quick-match teaser.
6. Recommendations	Recommendation worker, stored recommendations, dashboard, saved jobs.
7. AI Support	OpenRouter integration, cached explanations/extraction support, usage logs.
8. Emails and Privacy Polish	Email preferences, weekly digest, policy pages, deletion flows, error states.

28. Locked Product Decisions

- Product name: TuniTech Abroad.
- Candidate market: Tunisia.
- First destination market: France.
- Job source: France Travail API Offres d emploi.
- Future alternance source: La Bonne Alternance API.
- Primary UI language: French.
- Backend: Django.
- Frontend: Django templates + HTMX + Tailwind.
- Database: PostgreSQL.
- Background jobs: Redis + Celery/RQ.
- Auth: Google, GitHub, email/password.
- LLM: OpenRouter, controlled and cached.
- Scoring: rule-based fit score.
- Recommendations: included in MVP.
- Admin: Django Admin included in MVP.
- LinkedIn profile import: excluded from MVP.
- GDPR/privacy basics: included in MVP.

29. Next Document

After BRD and PRD review/validation, the next document is Technical Architecture v1. It should define Django app structure, database schema, background workers, job ingestion pipeline, skill taxonomy implementation, CV parsing pipeline, matching engine, recommendation engine, OpenRouter service, email service, admin implementation, privacy/data deletion flow, deployment plan, and testing strategy.

30. References Used for Planning

- France Travail API Offres d'emploi: <https://www.data.gouv.fr/dataservices/api-offres-emploi>
- La Bonne Alternance developer/API space: <https://labonnealternance.apprentissage.beta.gouv.fr/espace-developpeurs>
- CNIL recruitment and GDPR guide: <https://www.cnil.fr/fr/le-guide-du-recrutement>
- LinkedIn Profile API restriction note: <https://learn.microsoft.com/en-us/linkedin/shared/integrations/people/profile-api>